MATPLOTLIB LIBRARY

CLCoding

PYplot

```
plt.plot()
plt.scatter()
plt.bar()
plt.hist()
plt.pie()
plt.show()
```

FIGURE AND AXES

```
plt.subplots()
plt.figure()          ax.set.title()
ax.set.title(a        ax.set.xlabel()
ax.set.ylabel()       ax.set.ylabel()
ax.legend()           plt.axis()
```

COLORMAP

```
plt.cm.viridis
plt.cm.inferno
plt.cm.RdYiBu
plt.cm.gray
```

LINE

```
plt.plot(x, y)
    adiitiona l : color='red'
    linewdith=2  linestyle='–'
    marker='o: markersize=to
    markeredgecolor='black
    markerfacecolor='blue
```

MARKERS

```
plt.plot(x, y, marker='o')
plt.plot(x, y, marker='*')
plt.plot(x, y, marker='s')
```

IMAGES

```
plt.imshow(Z)
plt.imshow(Z.
cmap='gray)
plt.colorbar
```

BAR

```
plt.bar(x, y)
plt.bar(x, y,
width=0.5)
barh(x, y)
```

BAR

```
plt.bar(x, y)
plt.bar(x, y . width=0.5)
barhh(x, y)
```

STYLE

```
plt.style.use('classic')
plt.style.use('ggplot')
```

MATPLOTLIB CHEATSHEET

PYPLOT

- Create a figure and ax aplz.s›
- Plot data, ax.plot(x,y)
- Display a figure
- Plt.savefig('plot.png')

FIGURE

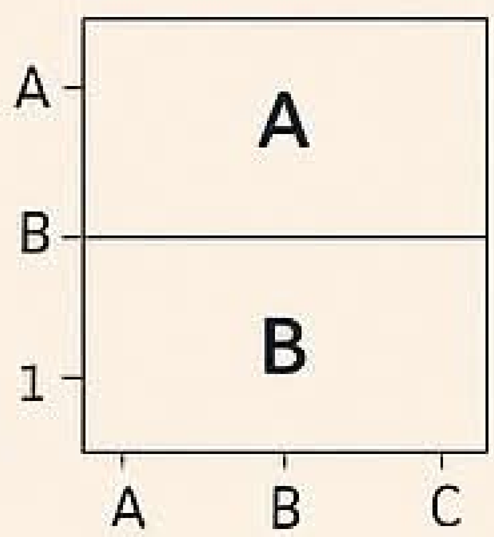
- Set size. plt.figure(figt z: 4))
- Adjust subplot parame-ters
  fig, subplats.adjust(left=0.1,
  right=0.9, torion,0,1, top,9)

SCATTER

- ax.scatter(x,y,s100, c='r',
  marker='x')

SUBPLOTS

fig . ax =
plt.sublots(
nrows=2, ncols1)



AXES

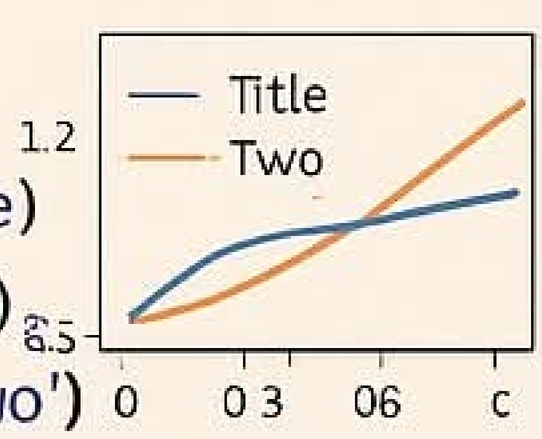
- Set labels(X)
- ax.set_ylabel('x')
- ax.set_title('Title')

TICKS

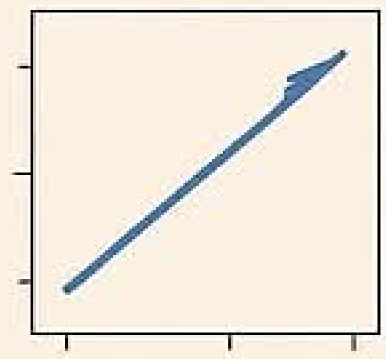
- ax.set_xicicks(1, 2, 3)
- ax.set_ticklabels[''a', 'b, 'c'])
- ax.set_xlim (0,10)
- ax.set_yscale('log')

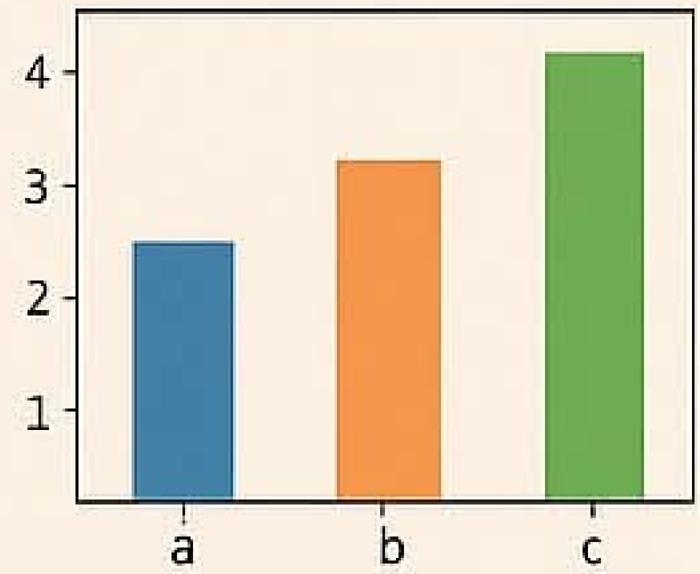
STYLE

- ax.set_title(Title)
- ax.set_ylabel('x')
- legend(['one, 'two')



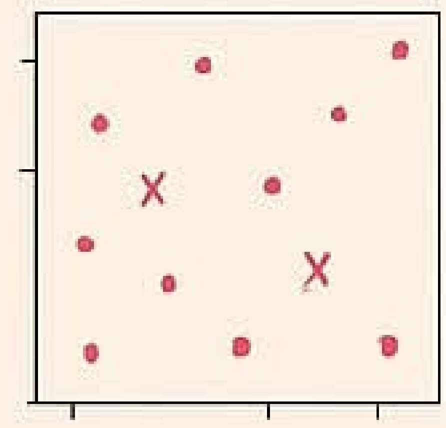
BAR CHAR





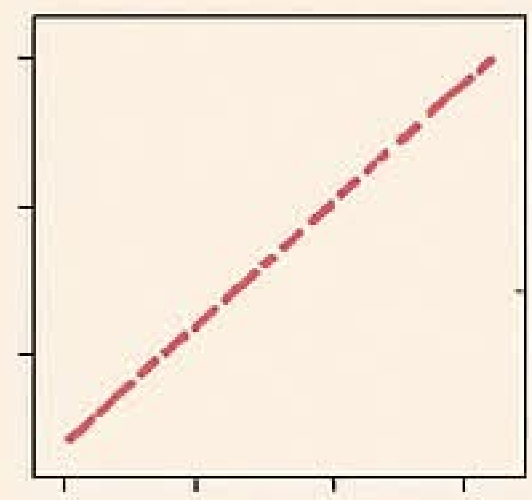
PLOT

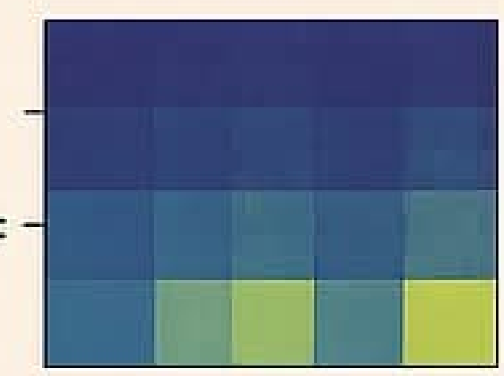
- ax.plot(x, y)
- ax.scatter(x,y)
- ax.bar(x, helght)



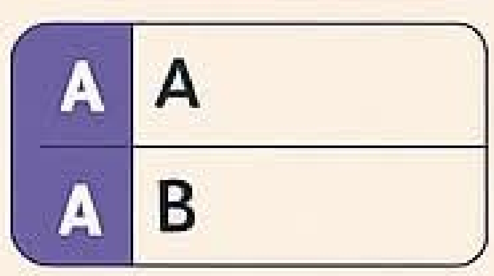
LINESTYLE

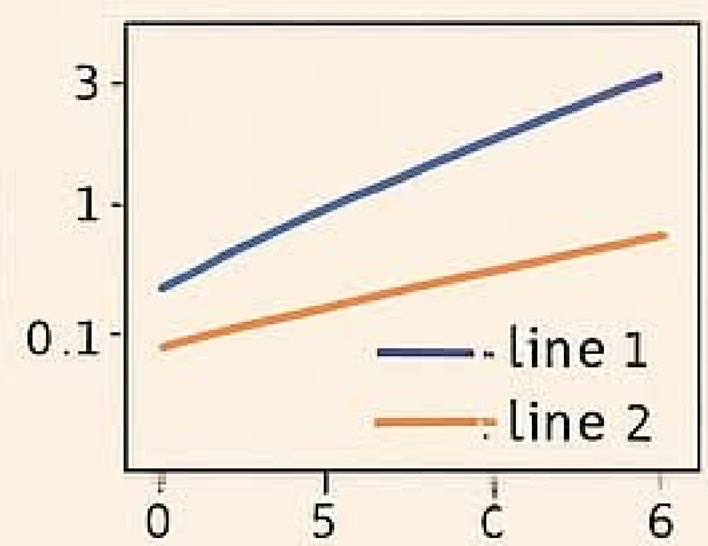
- ax.plot(x,.y,
  linestyle='solid')
- ax.plot(x, y,
  linestyle='dashed')
- ax.plot(x, y,
  linestyle='dotted')
- ax.plot(x.y, color=
  'r'. linewidth=2)





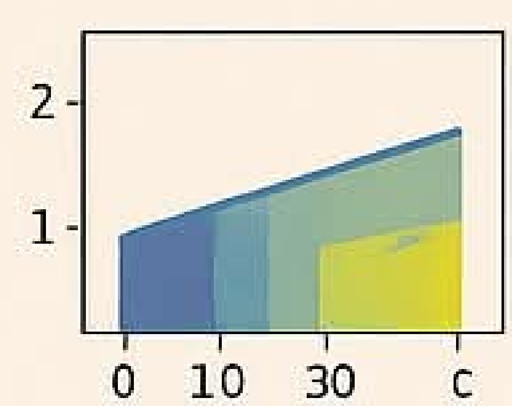
SUBPLOTS





AXES

- ax.set.xlabel('x')
- ax.set.yylabel('y')
- ax.set.title('Title')



TICKS

- ax.set.xtucks[1, 2, 3]
- ax.set.ticklabels('a', 'b, 'c')
- ax.set_ylim (0, 10)
- ax.set.yscale('log')

CLCoding

MATPLOTLIB LIBRARY

CLCoding

Importing

```
import matplotlb.pyplot
as plt
```

Creating a Plot

```
plt.plot([1,2,3],[4,5,6)
```

Customizing a Plot

```
plt.plot(x,y,color='red')
plt.ylot(x,y,axis label')
plt.plot(x,y,marker='o')
plt.plot(x,y,label='my data
```

Labels

```
plt.xlabel('x axis label')
plt.ylabel('y axis label')
plt.legend()
```

Grid

```
plt.grid(True)
```

Subplots

```
fig,ax = plt.subplots()
fig,(ax1, ax2) = plt.subplots)
```

Displaying the Plot

Creating a Figure

```
plt.figure()
```

Customizing a Plot

```
plt.plot(x,y,color='red')
plt.plot(x,y,yxis label')
plt.plot(x,y,marker='o')
plt.plot(x,y,label=my data)
```

Title

```
plt.title(Title of the Plot)
```

Axis Limits

```
plt.axis([xmin,xmax,
ymin,ymax])
```

Ticks

```
plt.xticks([1,2,3])
plt.yticks([4,5,6])
```

Subplots

```
fig,ax = plt.subplots()
fig,(ax1, ax2) =plt.subplots2)
```

Displaying the Plot

```
plt.show()
```

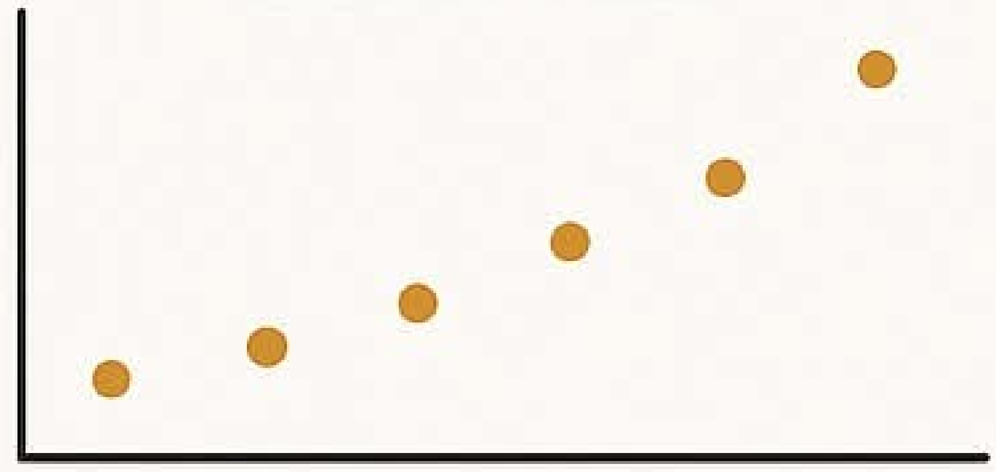
MATPLOTLIB CHEATSHEET

CLCoding

PLOT

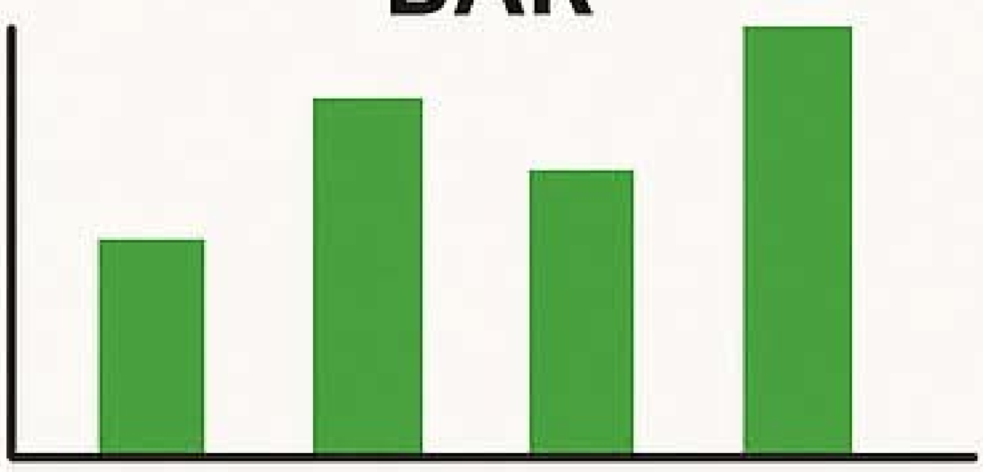
plt.plot(1, 2, 3, 4],[1,4, 9, 16])

SCATTER



plt.scatter(1, 2, 3, 4],[1, 4,9, 16)

BAR



plt.bar(['A', 'B', 'C', 'D'], (3,10, 5, 8)

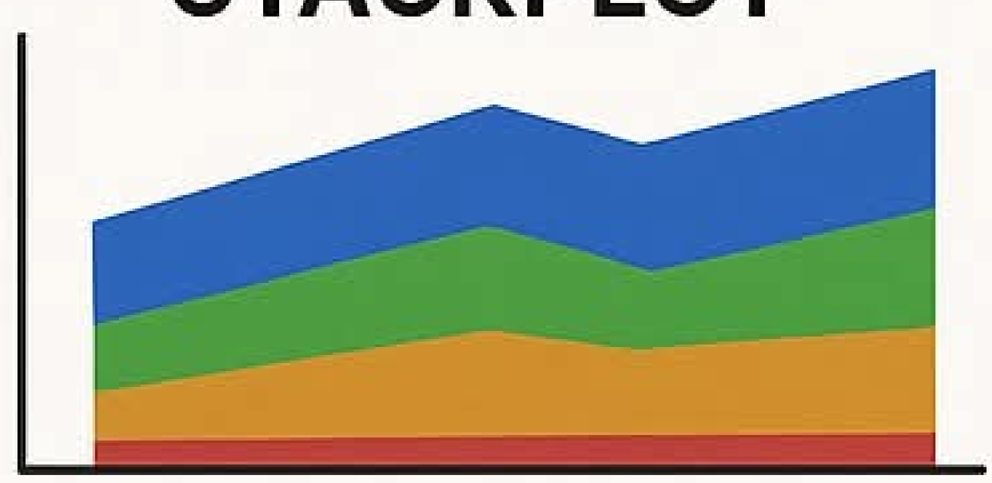
HISTOGRAM

plt.hist(data, bins=10)

SUBPLOTS

```
fig, ax = plt.subplots()
ax.plot(x, y)
ax.scatter(x,y)
```

STACKPLOT



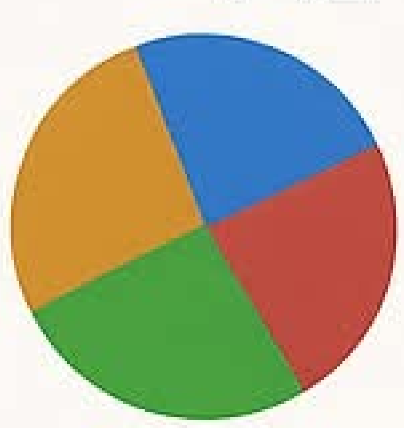
plt.stackplot(x, y1, y2, y3)

TABLE

A	B	C	C
1	4	7	7
2	5	8	8

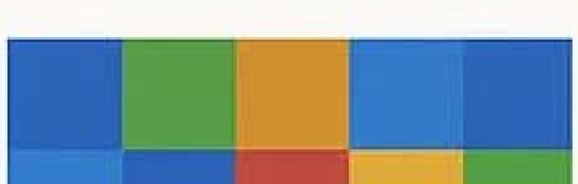
```
plt.table(cellText=vals,
      colLabels=cols)
```

PIE



plt.pie(1, 2, 3, 4)

IMSHOW



TABLE

A	B	C	C